

Technology Stack (Architecture & Stack) - ShopEZ Stock Trading Platform

Parameter	Details
Date	18 June 2026
Team ID	SE-09
Project Name	ShopEZ Stock Trading Platform
Maximum Marks	4 Marks

Technical Architecture

The Deliverable shall include the architectural diagram and the information as per table-1 & table-2.

The ShopEZ Stock Trading Platform leverages a modern 3-Tier Architecture designed for real-time responsiveness and high scalability:

- User Interface (Client Layer):** React-based single page application styled with a clean light/pastel palette, employing Chart.js for data visualization and Axios for RESTful API communication.
- Application Logic (Server Layer):** Node.js and Express RESTful API server handling authentication (JWT), trade validation logic, portfolio computations, and simulated live stock feed ticks.
- Data Storage (Database Layer):** MongoDB Atlas cloud database storing collections for users, stocks, portfolios, and transaction records.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	Responsive web interface for desktop and mobile browsers, styling using soft pastel design tokens.	HTML, CSS, JavaScript (React.js, Vite, Chart.js, Bootstrap)
2.	Application Logic-1	RESTful API server managing authentication, order validations, and transactions.	Node.js, Express framework
3.	Application Logic-2	Background Stock Price Simulator updating collection prices tick-by-tick.	JavaScript, Node.js background intervals
4.	Database	Local database storing user records, stock ticks, holdings, and transactions.	MongoDB

S.No	Component	Description	Technology
5.	Cloud Database	Cloud-hosted database cluster for high-availability production storage.	MongoDB Atlas
6.	File Storage	Configuration data, environmental variables, and static build outputs.	Local Filesystem
7.	External API-1	Initial stock listings seed data or live external market feed.	Finnhub / AlphaVantage API
8.	Infrastructure	Cloud deployment environment for concurrent frontend and backend processes.	Render Cloud Platform

Table-2: Application Characteristics